Symbols and Glossary

This glossary defines the Generalized Notation Notation (GNN) language constructs used to specify Active Inference generative models, together with the canonical Active Inference symbols those constructs carry. Definitions follow the GNN syntax specification and the discrete POMDP exemplars distributed with the repository, and they align with the standard discrete-state-space formulation of Active Inference (Smékal and Friedman 2023; Da Costa et al. 2020; Parr et al. 2022).

Language Constructs I

A GNN file is an ordered, UTF-8 Markdown document whose level-2 headers name required and optional sections. The strict schema validator enforces section order, declaration grammar, and connection syntax; the constructs below are the load-bearing sections a parser must recognize; they are catalogued in tbl. 1.

Language Constructs I

Table 1: GNN language constructs a conforming parser must recognize, with the meaning each carries.

Construct	Meaning
`## GNNSection`	Required short identifier for the model block (no spaces, e.g. `ActInfPOMDP`).
`## GNNVersionAndFlags`	Required version declaration (`GNN v1` or `GNN v1.1`) with optional flags.
`## ModelName`	Required human-readable title of the generative model.
`## StateSpaceBlock`	Required block declaring every variable and matrix as `NAME[dim, …, type=…]`, one per line.
`## Connections`	Required edge list relating state-space variables, expressing the model's factor graph.
`## ModelAnnotation`	Optional free-text description of the model's modalities, factors, and assumptions.
`## InitialParameterization`	Optional concrete numeric values for declared matrices and vectors.
`## ActInfOntologyAnnotation`	Optional bindings from each variable to a CamelCase Active Inference ontology term.
`## ModelParameters`	Optional key-value dimensions (e.g. `num_hidden_states`, `num_obs`, `num_actions`) consumed by code generators.
`## Time`	Optional dynamics declaration: a time variable plus `Dynamic`/`Static`, `Discrete`/`Continuous`, and `ModelTimeHorizon`.
`NAME[d ,d ,…,type=…]`	A variable or tensor declaration; dimensions are positive integers or named references, and a `type` (`float`, `int`, `bool`) is required.
`A>B`	Directed (causal) connection operator: edge from `A` to `B`, e.g. `D>s` (a prior conditions a hidden state).
`A-B`	Undirected (bidirectional) connection operator, e.g. `s-A` (a hidden state participates in the likelihood mapping).
`A>B:label` / `A-B:label`	A v1.1 annotated edge; the trailing label documents the relation and is preserved but may be ignored for structural validation.

Language Constructs II

`default=...`	A v1.1 declaration hint (`uniform`, `zeros`, `ones`, `eye`, `random`) supplying an initialization for a matrix or vector.

The exemplar discrete POMDP agent declares the standard generative-model components of Active Inference over a discrete state space, mapping each GNN variable to its probabilistic meaning (Da Costa et al. 2020; Smith et al. 2022; Parr et al. 2022). Each row of tbl. 2 names the equation in sec. 3 that fixes the symbol's role, so the glossary and the formal statement cannot drift apart.

Active Inference Symbols I

Table 2: Active Inference symbols carried by a GNN specification, each bound to the equation in sec. 3 that defines it.

Symbol	Meaning
`A`	Likelihood (observation) matrix encoding $P(o \mid s)$, mapping hidden states to observation outcomes (eq. 2).
`B`	Transition matrix encoding $P(s' \mid s, u)$, mapping a previous state and action to the next state (eq. 3).
`C`	Preference vector: log-preferences over observation outcomes that bias the agent toward preferred outcomes (eq. 4).
`D`	Prior vector over initial hidden states (eq. 5).
`E`	Habit vector: an initial policy prior (baseline preference) over actions, entering the policy posterior (eq. 8).
`s`	Current hidden-state distribution; `s_prime` (`s'`) is the next hidden-state distribution.
`o`	Current observation, an integer index over outcome modalities.
π	Policy: a distribution over actions inferred from expected free energy (eq. 8).
`u`	The selected (sampled) action, sampled from the policy posterior.
`F`	Variational free energy, minimized during state inference to update beliefs from observations (eq. 6) (Friston 2010).
`G`	Expected free energy per policy, minimized during policy inference to score candidate actions (eq. 7) (Da Costa et al. 2020).
`t`	Discrete time step; the horizon T bounds the product in eq. 1.

The `## ActInfOntologyAnnotation` section binds each variable to a canonical term (`A=LikelihoodMatrix`, `B=TransitionMatrix`, `C=LogPreferenceVector`, `D=PriorOverHiddenStates`, `s=HiddenState`, `o=Observation`, π`=PolicyVector`), which downstream pipeline steps use for semantic analysis and validation. The same GNN specification feeds the project's rendering backends — including the 9 executable targets (PyMDP, RxInfer.jl, ActiveInference.jl, JAX, DisCoPy, PyTorch, NumPyro, Stan, bnlearn) — so that a model written once in this notation can be parsed, visualized, and executed across the 25-step pipeline (0–24) without restating its mathematics (Smékal and Friedman 2023; Heins et al. 2022; Felice et al. 2021).